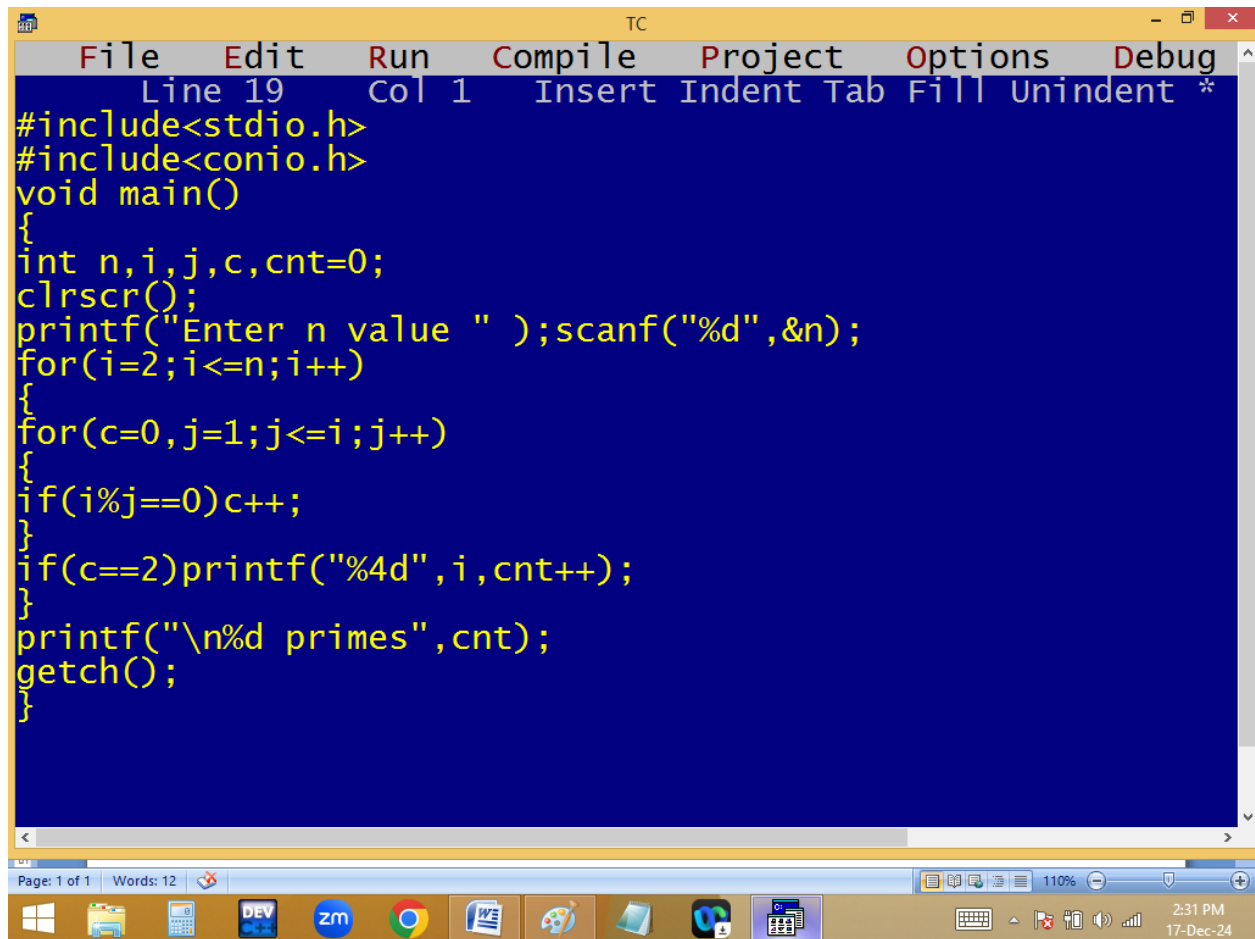


Finding 1..n primes and count.



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 19", "Col 1", and "Insert Indent Tab Fill Unindent *". The main editing area has a dark blue background with yellow text. The code is a C program to find prime numbers up to a given value 'n'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,c,cnt=0;
clrscr();
printf("Enter n value " );scanf("%d",&n);
for(i=2;i<=n;i++)
{
for(c=0,j=1;j<=i;j++)
{
if(i%j==0)c++;
}
if(c==2)printf("%4d",i,cnt++);
}
printf("\n%d primes",cnt);
getch();
}
```

At the bottom of the window, there is a status bar showing "Page: 1 of 1" and "Words: 12". Below the status bar is a taskbar with various icons, including Windows, File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, and others. The system clock in the bottom right corner shows "2:31 PM" and "17-Dec-24".

```

Enter n value 20
2 3 5 7 11 13 17 19
8 primes_

```

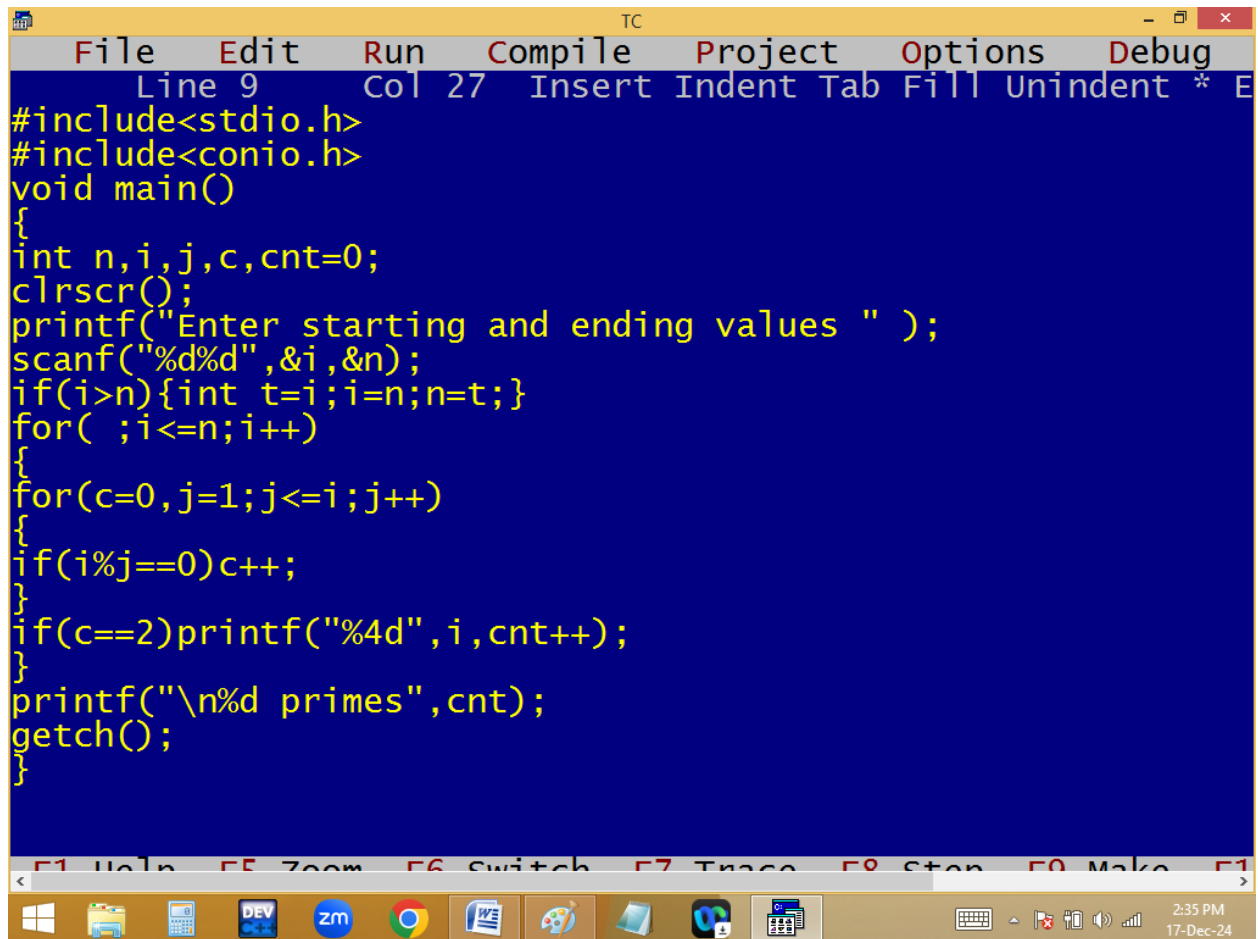
```

cnt=0; ✓
for(i=2; i<=n; i++)
{
  for(c=0, j=1; j<=i; j++)
  {
    if(i%j==0) c++;
  }
  if( c==2 ) p(i, cnt++);
}
p(cnt);

```

$\frac{n}{i}$	$\frac{i}{j}$	$\frac{j}{c}$	$\frac{cnt}{}$
10	✓ 2 % 1 = 0	0 1 2	0
	2 % 2 = 0		1
✓ 3 % 1 = 0		0 1 2	2
3 % 3 = 0		0 1 2 3	
4 % 1 = 0			
2 % 2 = 0			
4 % 4 = 0			

Finding n to n primes and count



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9", "Col 27", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text containing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,c,cnt=0;
clrscr();
printf("Enter starting and ending values " );
scanf("%d%d",&i,&n);
if(i>n){int t=i;i=n;n=t;}
for( ;i<=n;i++)
{
for(c=0,j=1;j<=i;j++)
{
if(i%j==0)c++;
}
if(c==2)printf("%4d",i,cnt++);
}
printf("\n%d primes",cnt);
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a status bar showing the time "2:35 PM" and the date "17-Dec-24".

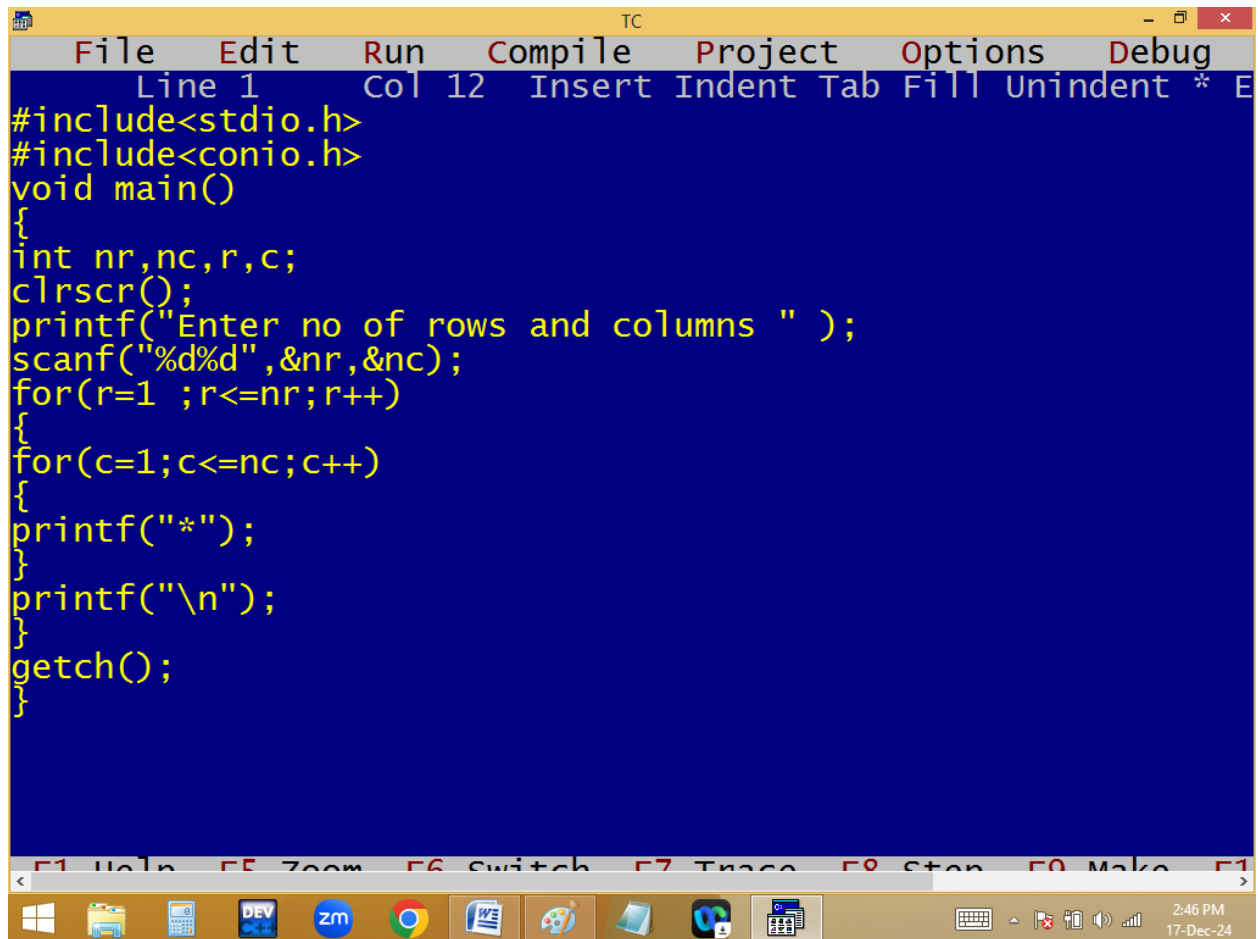
```
TC
Enter starting and ending values 200 100
101 103 107 109 113 127 131 137 139 149 151 157 163 167 17
199
21 primes_
```

Patterns:

* * * *

* * * *

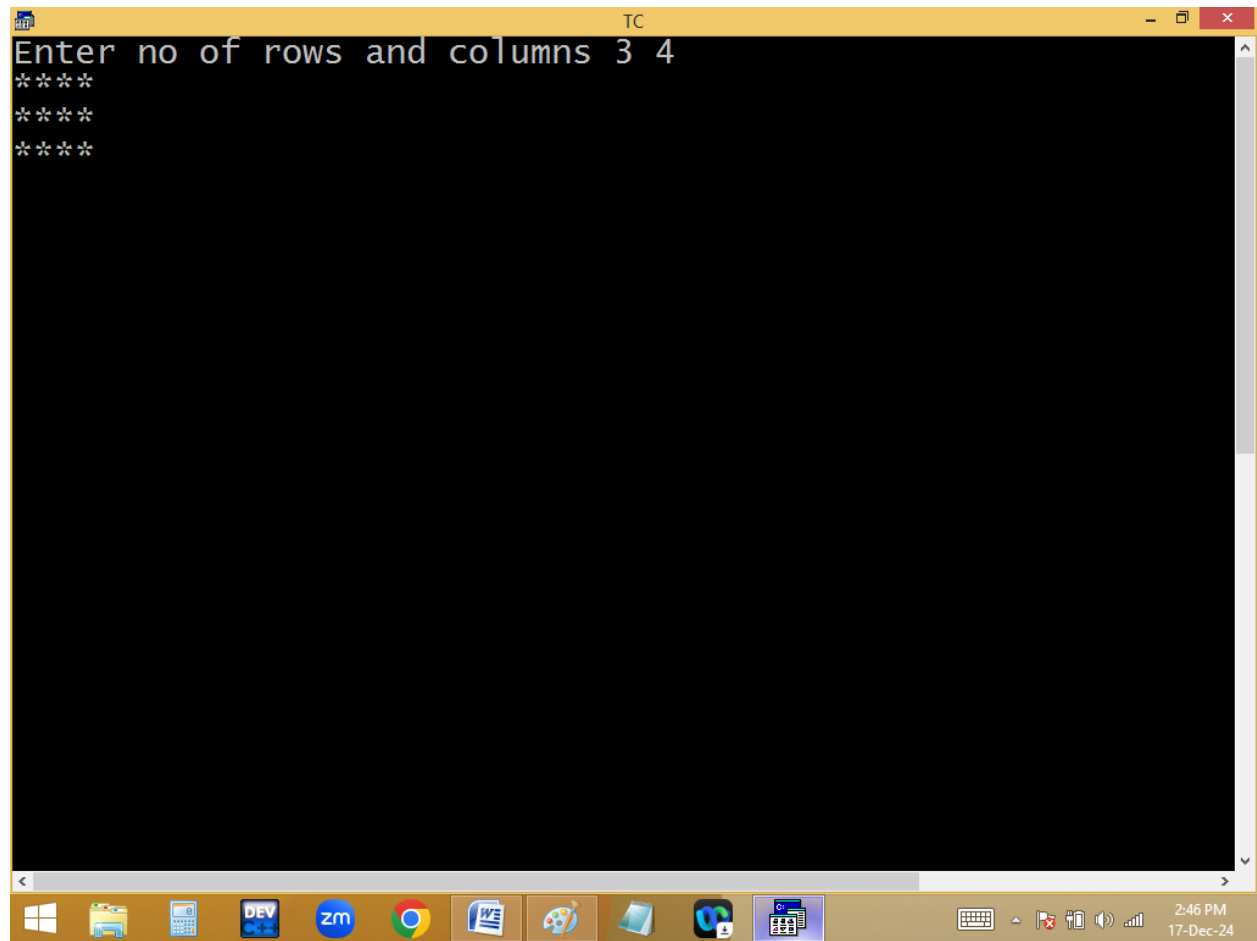
* * * *

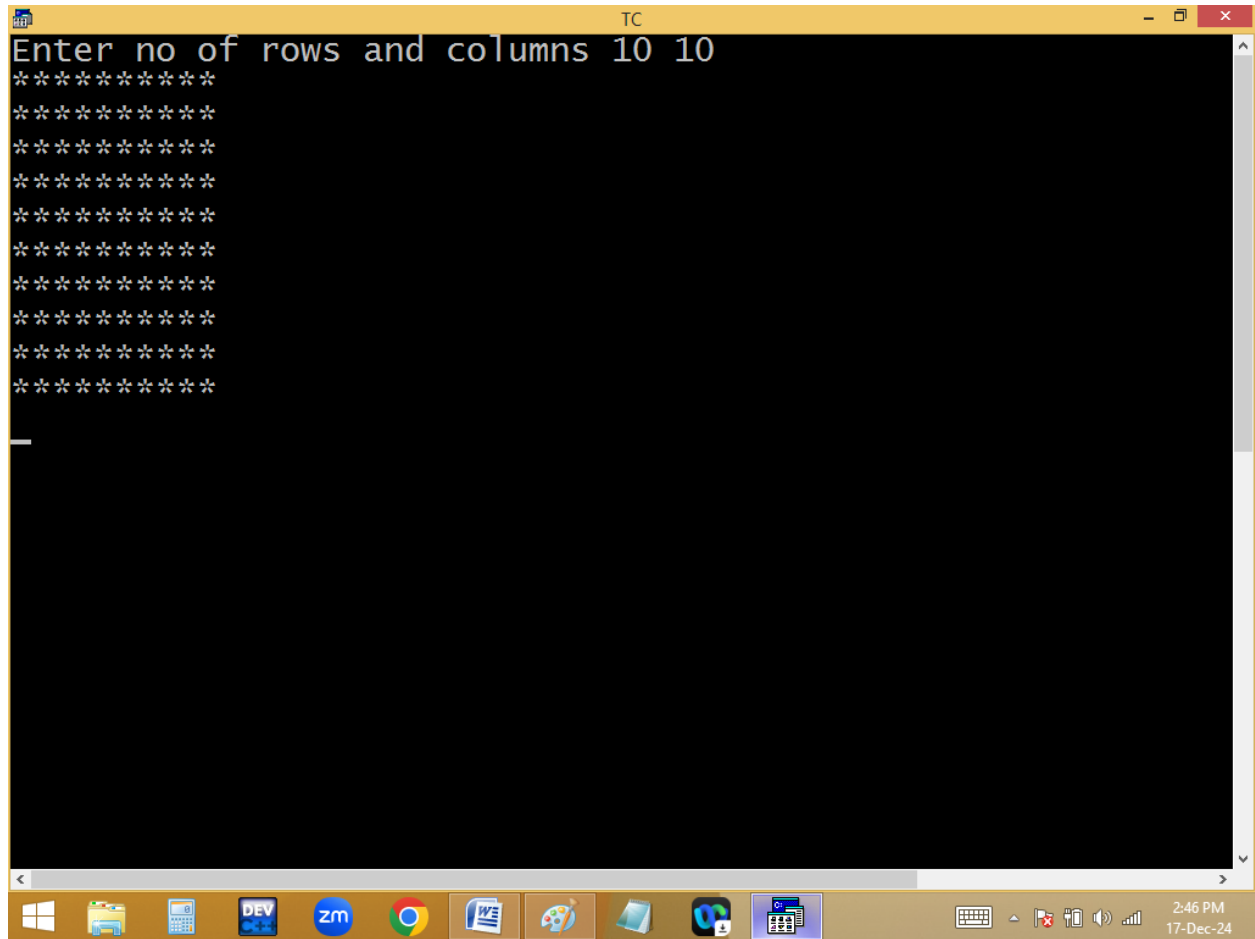


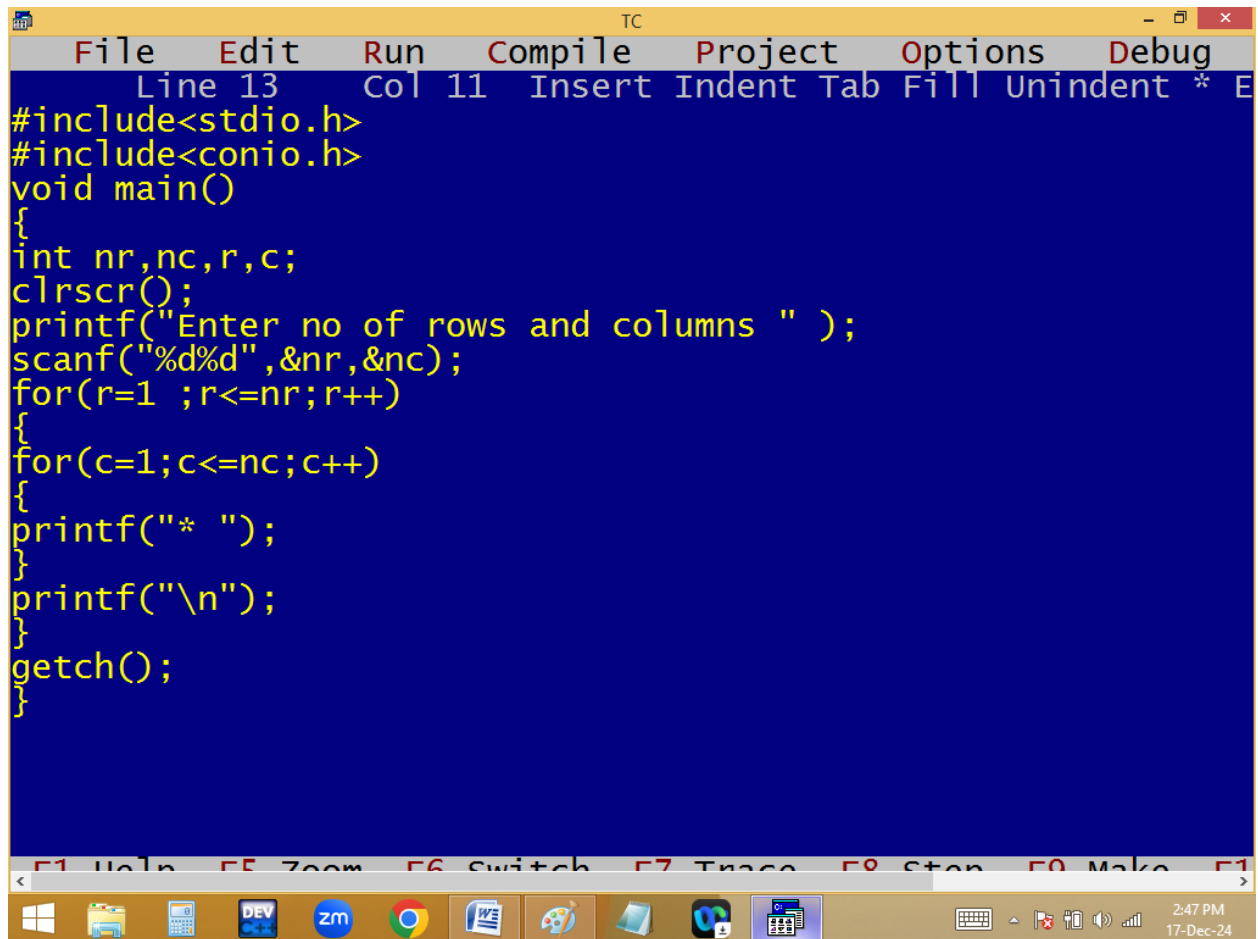
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 1" and "Col 12". The main editing area has a blue background with yellow text. The code is a C program that prompts the user for the number of rows and columns, then prints a grid of asterisks. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("*");
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:46 PM" and the date "17-Dec-24".



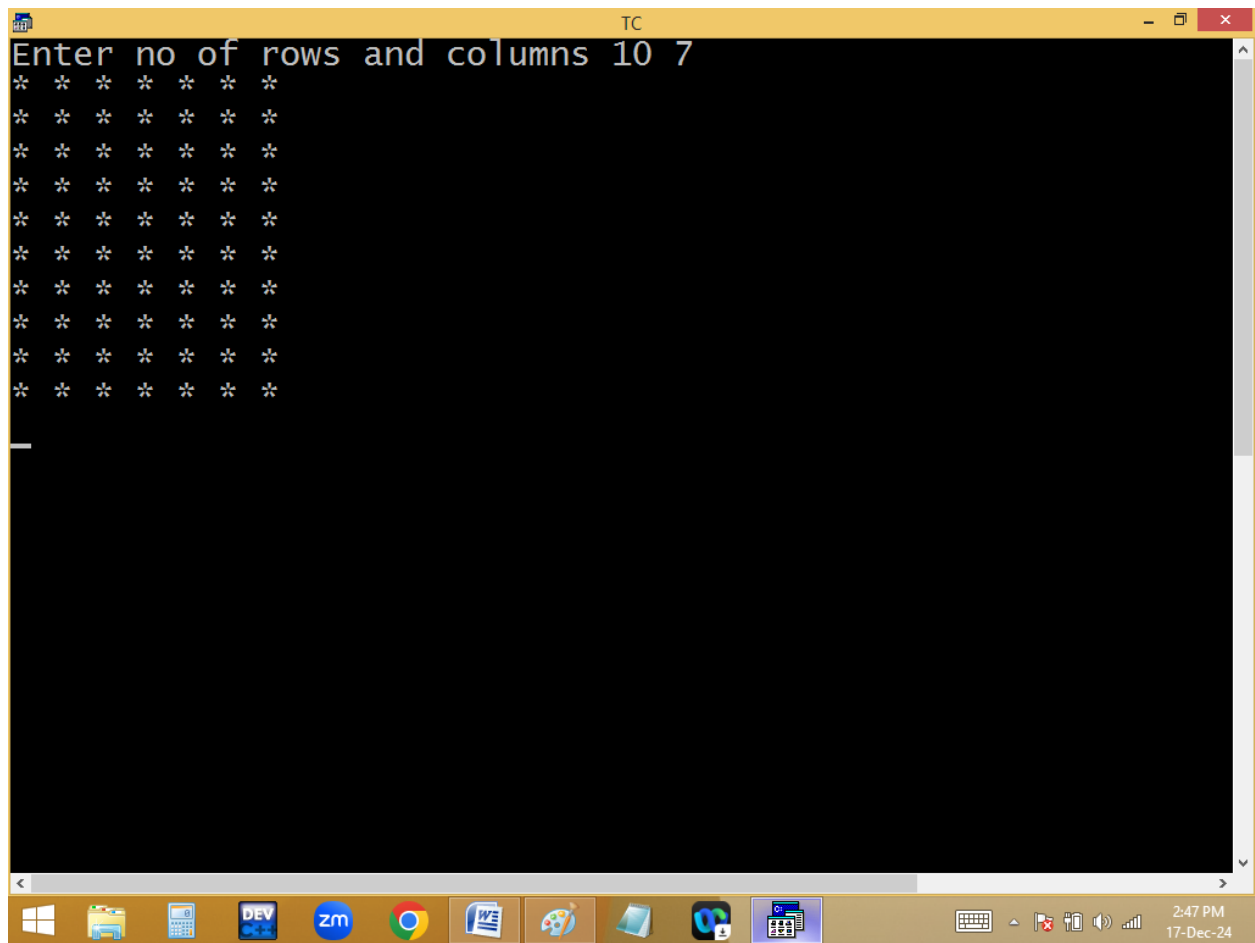


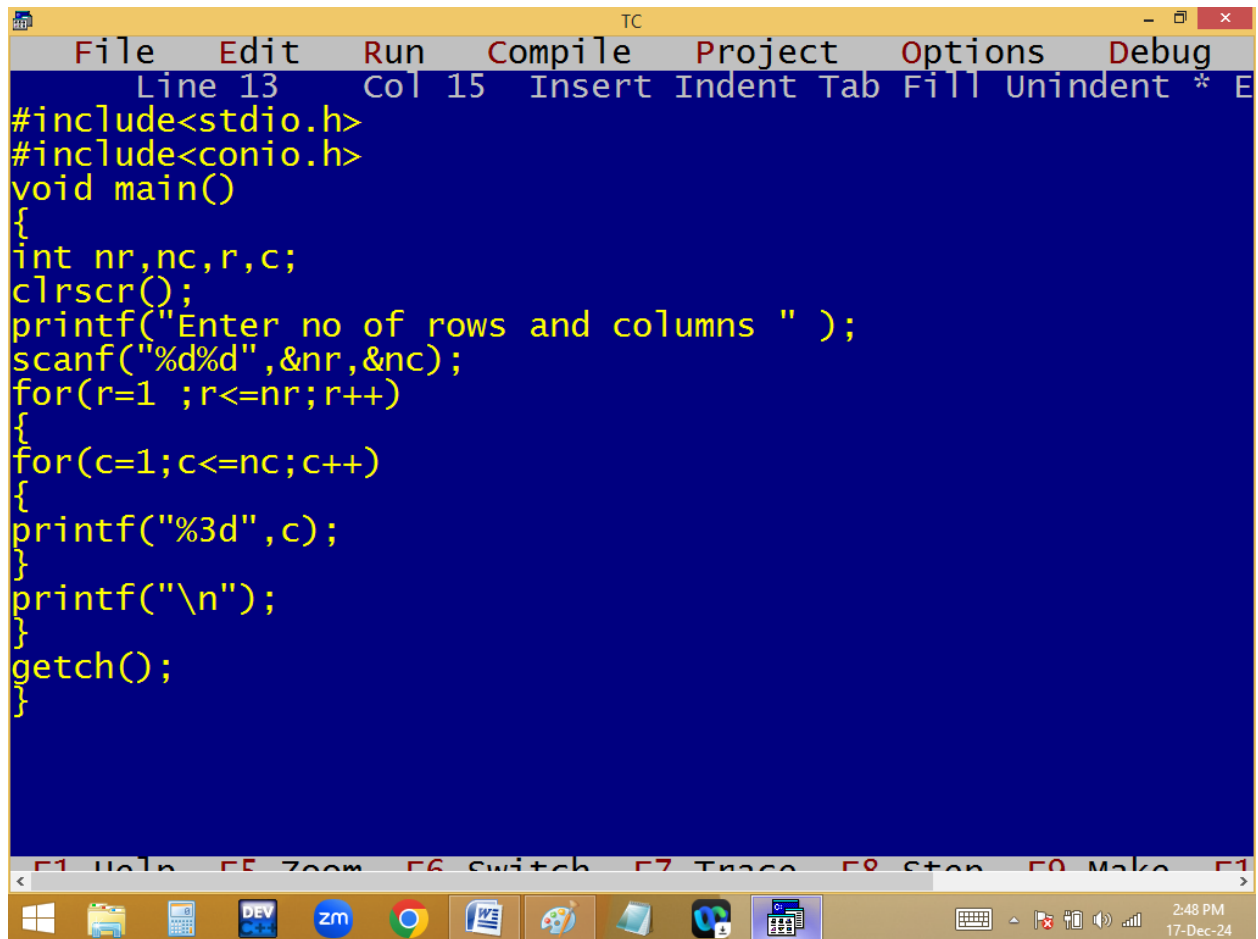


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 11", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and a multiplier "* E". The main editing area has a blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of asterisks. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns " );
    scanf("%d%d",&nr,&nc);
    for(r=1 ;r<=nr;r++)
    {
        for(c=1;c<=nc;c++)
        {
            printf("* ");
        }
        printf("\n");
    }
    getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:47 PM" and the date "17-Dec-24".



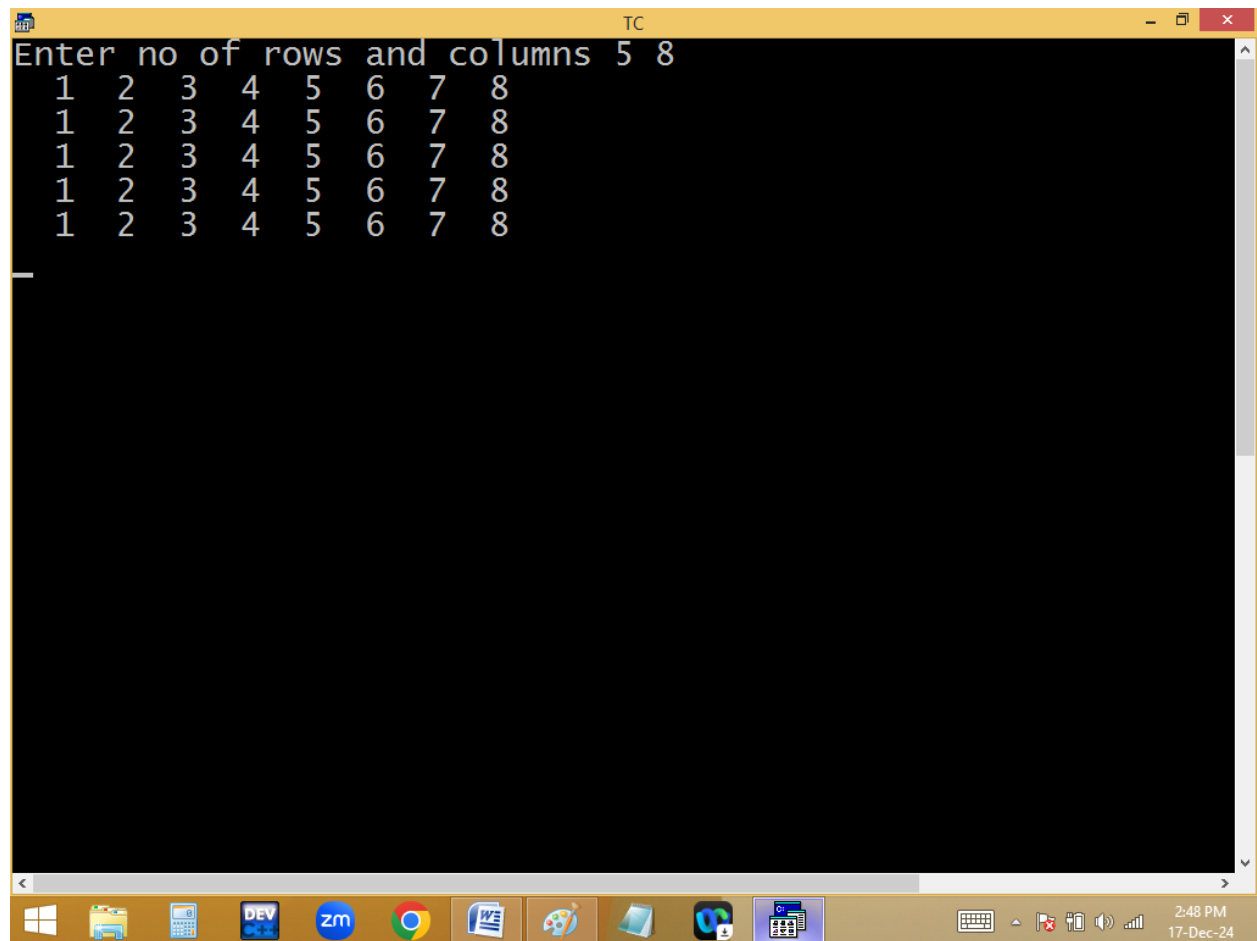


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 15", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of numbers from 1 to the total number of elements. The code is as follows:

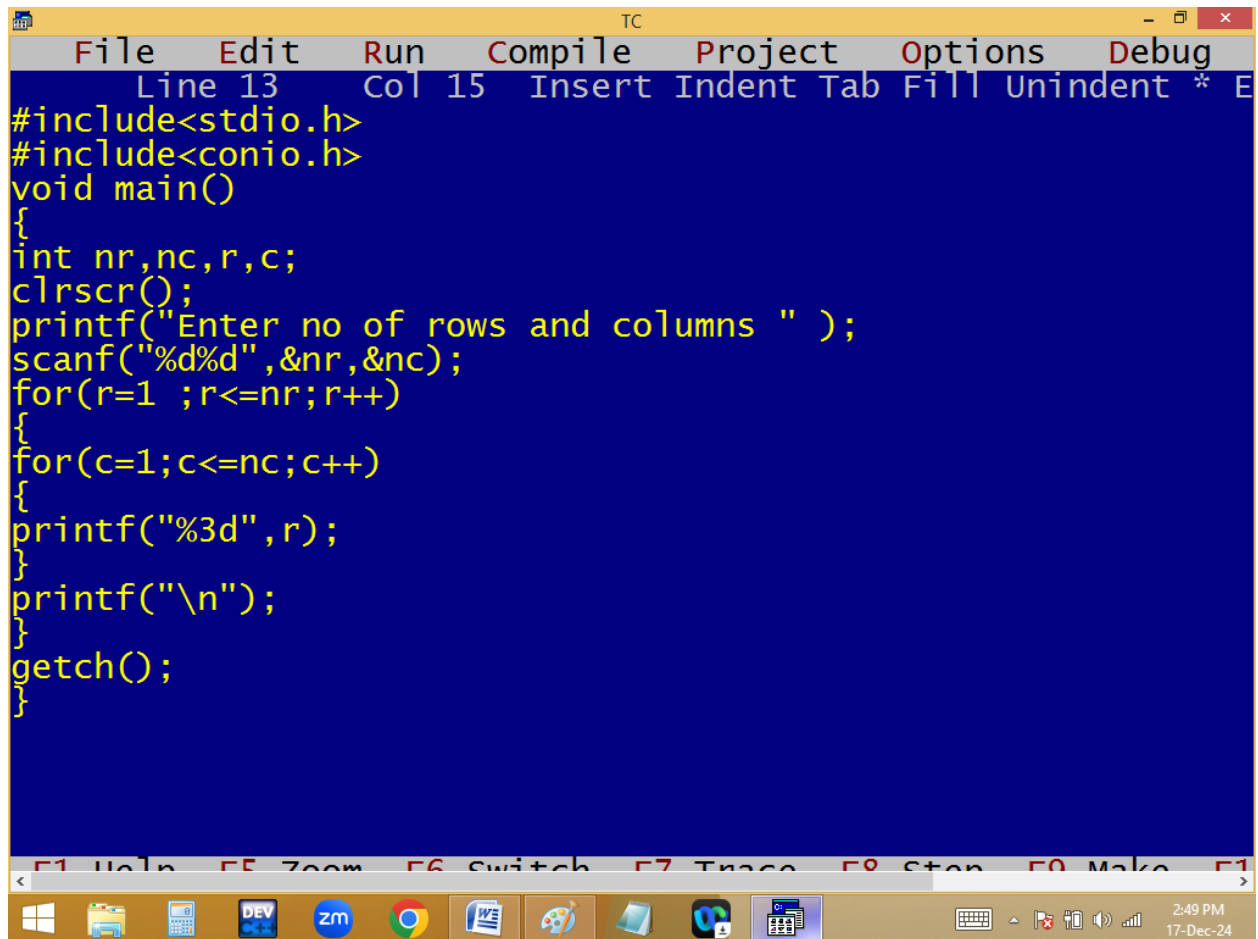
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns " );
    scanf("%d%d",&nr,&nc);
    for(r=1 ;r<=nr;r++)
    {
        for(c=1;c<=nc;c++)
        {
            printf("%3d",c);
        }
        printf("\n");
    }
    getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:48 PM" and the date "17-Dec-24".

```
TC
Enter no of rows and columns 5 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
```



The image shows a Windows desktop environment. A terminal window titled 'TC' is open, displaying a prompt 'Enter no of rows and columns 5 8' and a 5x8 grid of numbers from 1 to 8. The taskbar at the bottom contains icons for Windows, File Explorer, Calculator, DEV, ZM, Chrome, Word, a game controller, a folder, a game, and a calendar. The system tray on the right shows the time as 2:48 PM on 17-Dec-24.



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 15", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of numbers. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns " );
    scanf("%d%d",&nr,&nc);
    for(r=1 ;r<=nr;r++)
    {
        for(c=1;c<=nc;c++)
        {
            printf("%3d",r);
        }
        printf("\n");
    }
    getch();
}
```

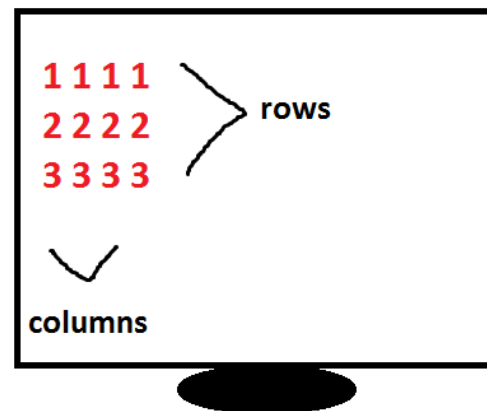
At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:49 PM" and the date "17-Dec-24".

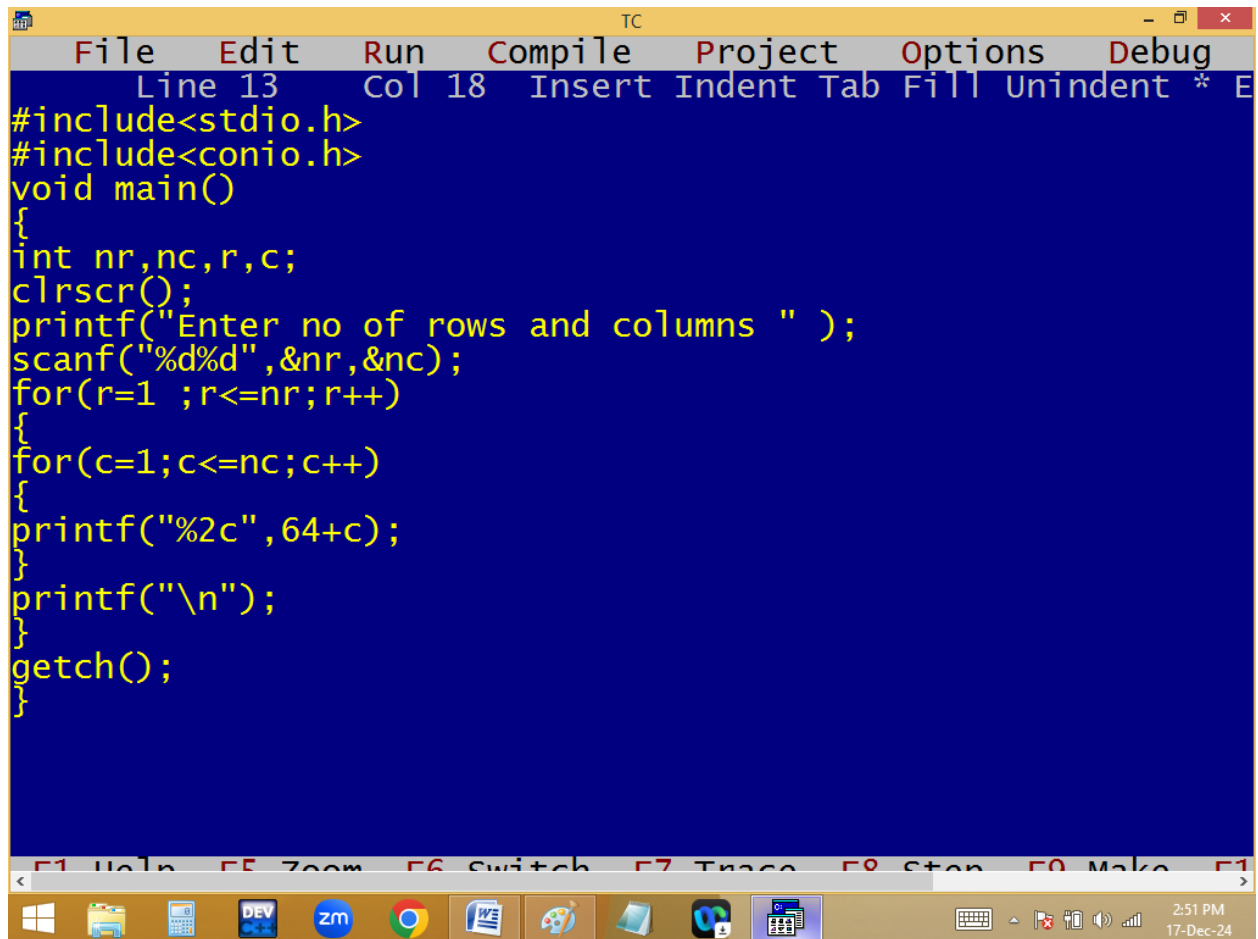
```
TC
Enter no of rows and columns 5 5
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

```
for( r=1; r<=3; r++)
{
  for( c=1; c<=4; c++)
  {
    p( r );
  }
  p("\n");
}
```

Handwritten annotations:

- Under r in the first loop: $\frac{r}{c}$
- Under c in the second loop: $\frac{r}{c}$
- Next to the first loop: $\frac{r}{c}$
- Next to the second loop: $\frac{r}{c}$
- Next to the third loop: $\frac{r}{c}$
- Next to the fourth loop: $\frac{r}{c}$
- Next to the fifth loop: $\frac{r}{c}$
- Next to the sixth loop: $\frac{r}{c}$
- Next to the seventh loop: $\frac{r}{c}$
- Next to the eighth loop: $\frac{r}{c}$
- Next to the ninth loop: $\frac{r}{c}$
- Next to the tenth loop: $\frac{r}{c}$
- Next to the eleventh loop: $\frac{r}{c}$
- Next to the twelfth loop: $\frac{r}{c}$
- Next to the thirteenth loop: $\frac{r}{c}$
- Next to the fourteenth loop: $\frac{r}{c}$
- Next to the fifteenth loop: $\frac{r}{c}$
- Next to the sixteenth loop: $\frac{r}{c}$
- Next to the seventeenth loop: $\frac{r}{c}$
- Next to the eighteenth loop: $\frac{r}{c}$
- Next to the nineteenth loop: $\frac{r}{c}$
- Next to the twentieth loop: $\frac{r}{c}$
- Next to the twenty-first loop: $\frac{r}{c}$
- Next to the twenty-second loop: $\frac{r}{c}$
- Next to the twenty-third loop: $\frac{r}{c}$
- Next to the twenty-fourth loop: $\frac{r}{c}$
- Next to the twenty-fifth loop: $\frac{r}{c}$
- Next to the twenty-sixth loop: $\frac{r}{c}$
- Next to the twenty-seventh loop: $\frac{r}{c}$
- Next to the twenty-eighth loop: $\frac{r}{c}$
- Next to the twenty-ninth loop: $\frac{r}{c}$
- Next to the thirtieth loop: $\frac{r}{c}$
- Next to the thirty-first loop: $\frac{r}{c}$
- Next to the thirty-second loop: $\frac{r}{c}$
- Next to the thirty-third loop: $\frac{r}{c}$
- Next to the thirty-fourth loop: $\frac{r}{c}$
- Next to the thirty-fifth loop: $\frac{r}{c}$
- Next to the thirty-sixth loop: $\frac{r}{c}$
- Next to the thirty-seventh loop: $\frac{r}{c}$
- Next to the thirty-eighth loop: $\frac{r}{c}$
- Next to the thirty-ninth loop: $\frac{r}{c}$
- Next to the fortieth loop: $\frac{r}{c}$
- Next to the forty-first loop: $\frac{r}{c}$
- Next to the forty-second loop: $\frac{r}{c}$
- Next to the forty-third loop: $\frac{r}{c}$
- Next to the forty-fourth loop: $\frac{r}{c}$
- Next to the forty-fifth loop: $\frac{r}{c}$
- Next to the forty-sixth loop: $\frac{r}{c}$
- Next to the forty-seventh loop: $\frac{r}{c}$
- Next to the forty-eighth loop: $\frac{r}{c}$
- Next to the forty-ninth loop: $\frac{r}{c}$
- Next to the fiftieth loop: $\frac{r}{c}$
- Next to the fifty-first loop: $\frac{r}{c}$
- Next to the fifty-second loop: $\frac{r}{c}$
- Next to the fifty-third loop: $\frac{r}{c}$
- Next to the fifty-fourth loop: $\frac{r}{c}$
- Next to the fifty-fifth loop: $\frac{r}{c}$
- Next to the fifty-sixth loop: $\frac{r}{c}$
- Next to the fifty-seventh loop: $\frac{r}{c}$
- Next to the fifty-eighth loop: $\frac{r}{c}$
- Next to the fifty-ninth loop: $\frac{r}{c}$
- Next to the sixtieth loop: $\frac{r}{c}$
- Next to the sixty-first loop: $\frac{r}{c}$
- Next to the sixty-second loop: $\frac{r}{c}$
- Next to the sixty-third loop: $\frac{r}{c}$
- Next to the sixty-fourth loop: $\frac{r}{c}$
- Next to the sixty-fifth loop: $\frac{r}{c}$
- Next to the sixty-sixth loop: $\frac{r}{c}$
- Next to the sixty-seventh loop: $\frac{r}{c}$
- Next to the sixty-eighth loop: $\frac{r}{c}$
- Next to the sixty-ninth loop: $\frac{r}{c}$
- Next to the seventieth loop: $\frac{r}{c}$
- Next to the seventy-first loop: $\frac{r}{c}$
- Next to the seventy-second loop: $\frac{r}{c}$
- Next to the seventy-third loop: $\frac{r}{c}$
- Next to the seventy-fourth loop: $\frac{r}{c}$
- Next to the seventy-fifth loop: $\frac{r}{c}$
- Next to the seventy-sixth loop: $\frac{r}{c}$
- Next to the seventy-seventh loop: $\frac{r}{c}$
- Next to the seventy-eighth loop: $\frac{r}{c}$
- Next to the seventy-ninth loop: $\frac{r}{c}$
- Next to the eightieth loop: $\frac{r}{c}$
- Next to the eighty-first loop: $\frac{r}{c}$
- Next to the eighty-second loop: $\frac{r}{c}$
- Next to the eighty-third loop: $\frac{r}{c}$
- Next to the eighty-fourth loop: $\frac{r}{c}$
- Next to the eighty-fifth loop: $\frac{r}{c}$
- Next to the eighty-sixth loop: $\frac{r}{c}$
- Next to the eighty-seventh loop: $\frac{r}{c}$
- Next to the eighty-eighth loop: $\frac{r}{c}$
- Next to the eighty-ninth loop: $\frac{r}{c}$
- Next to the ninetieth loop: $\frac{r}{c}$
- Next to the ninety-first loop: $\frac{r}{c}$
- Next to the ninety-second loop: $\frac{r}{c}$
- Next to the ninety-third loop: $\frac{r}{c}$
- Next to the ninety-fourth loop: $\frac{r}{c}$
- Next to the ninety-fifth loop: $\frac{r}{c}$
- Next to the ninety-sixth loop: $\frac{r}{c}$
- Next to the ninety-seventh loop: $\frac{r}{c}$
- Next to the ninety-eighth loop: $\frac{r}{c}$
- Next to the ninety-ninth loop: $\frac{r}{c}$
- Next to the hundredth loop: $\frac{r}{c}$





The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 18", and a list of editing commands: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

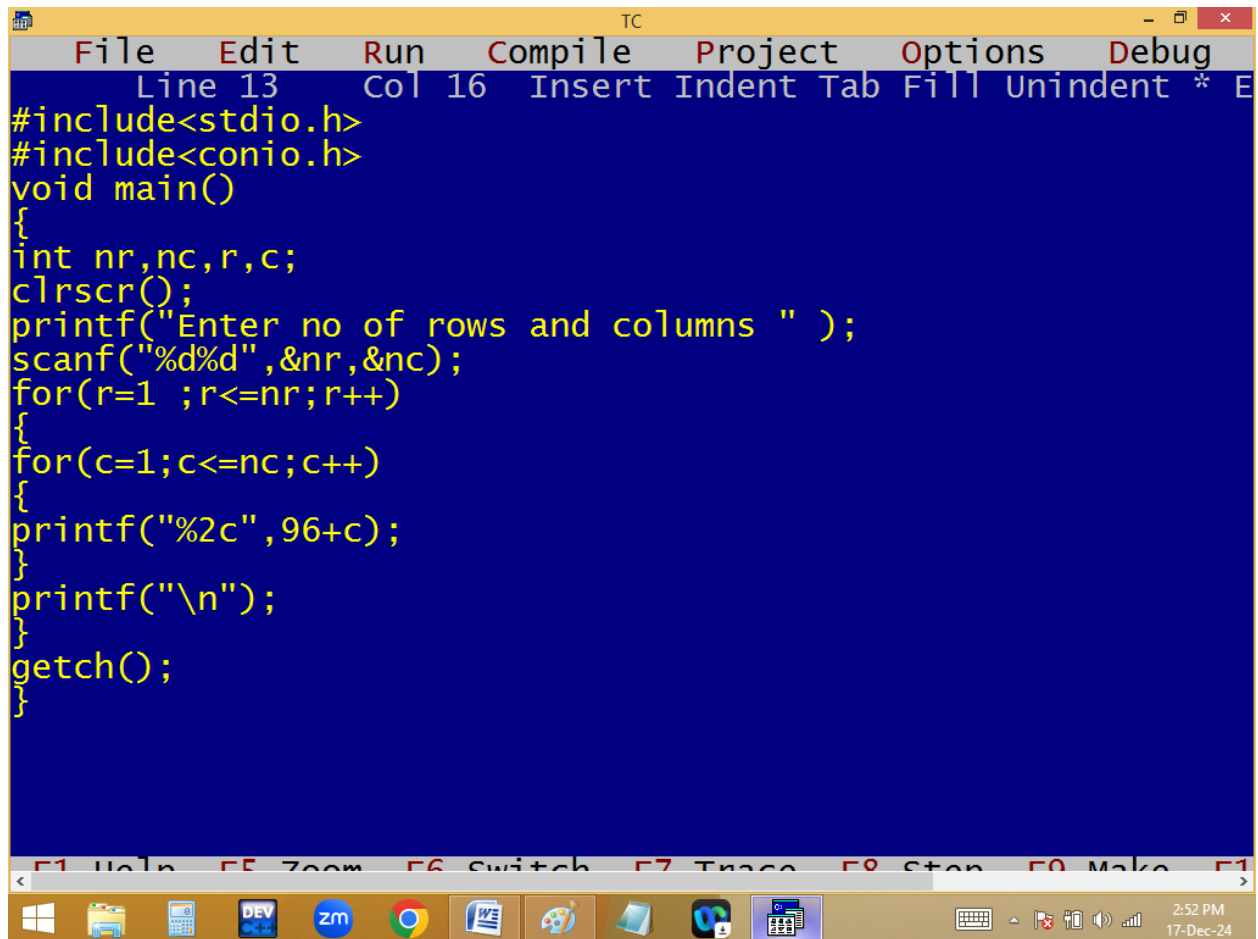
At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:51 PM" and the date "17-Dec-24".

```
TC
```

```
Enter no of rows and columns 10 26  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
```

```
<
```

2:51 PM
17-Dec-24



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 16", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of asterisks. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns " );
    scanf("%d%d",&nr,&nc);
    for(r=1 ;r<=nr;r++)
    {
        for(c=1;c<=nc;c++)
        {
            printf("%2c",96+c);
        }
        printf("\n");
    }
    getch();
}
```

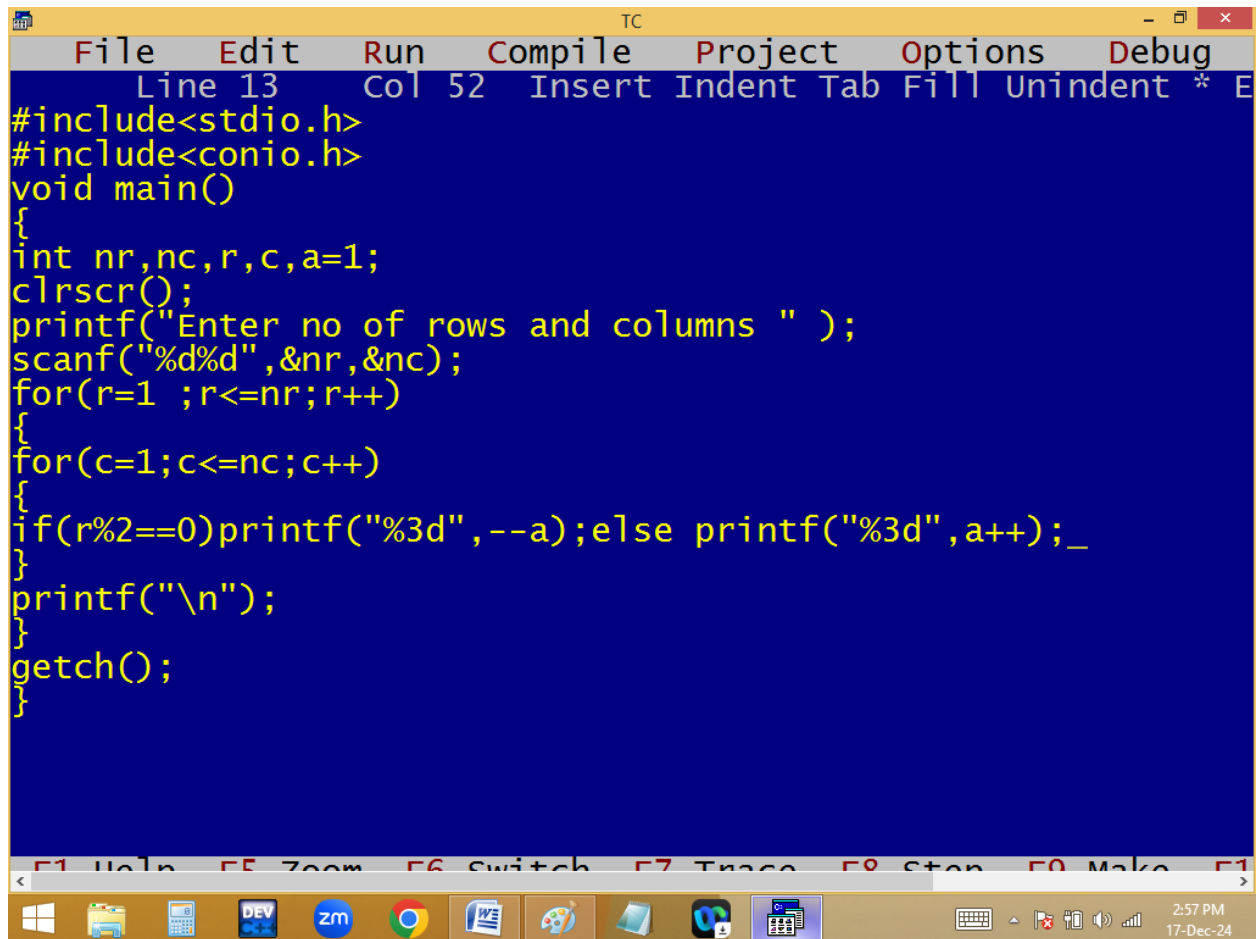
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "2:52 PM" and "17-Dec-24".

TC

Enter no of rows and columns 10 26

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z

Windows taskbar icons: File Explorer, Calculator, DEV, zm, Google Chrome, Word, Paint, Edge, Outlook, Excel. System clock: 2:52 PM, 17-Dec-24.



```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 52 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%3d",--a);else printf("%3d",a++);_
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:57 PM
17-Dec-24

```

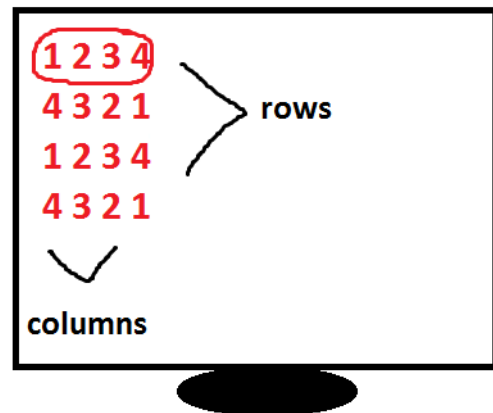
TC
Enter no of rows and columns 4 4
1 2 3 4
4 3 2 1
1 2 3 4
4 3 2 1

```

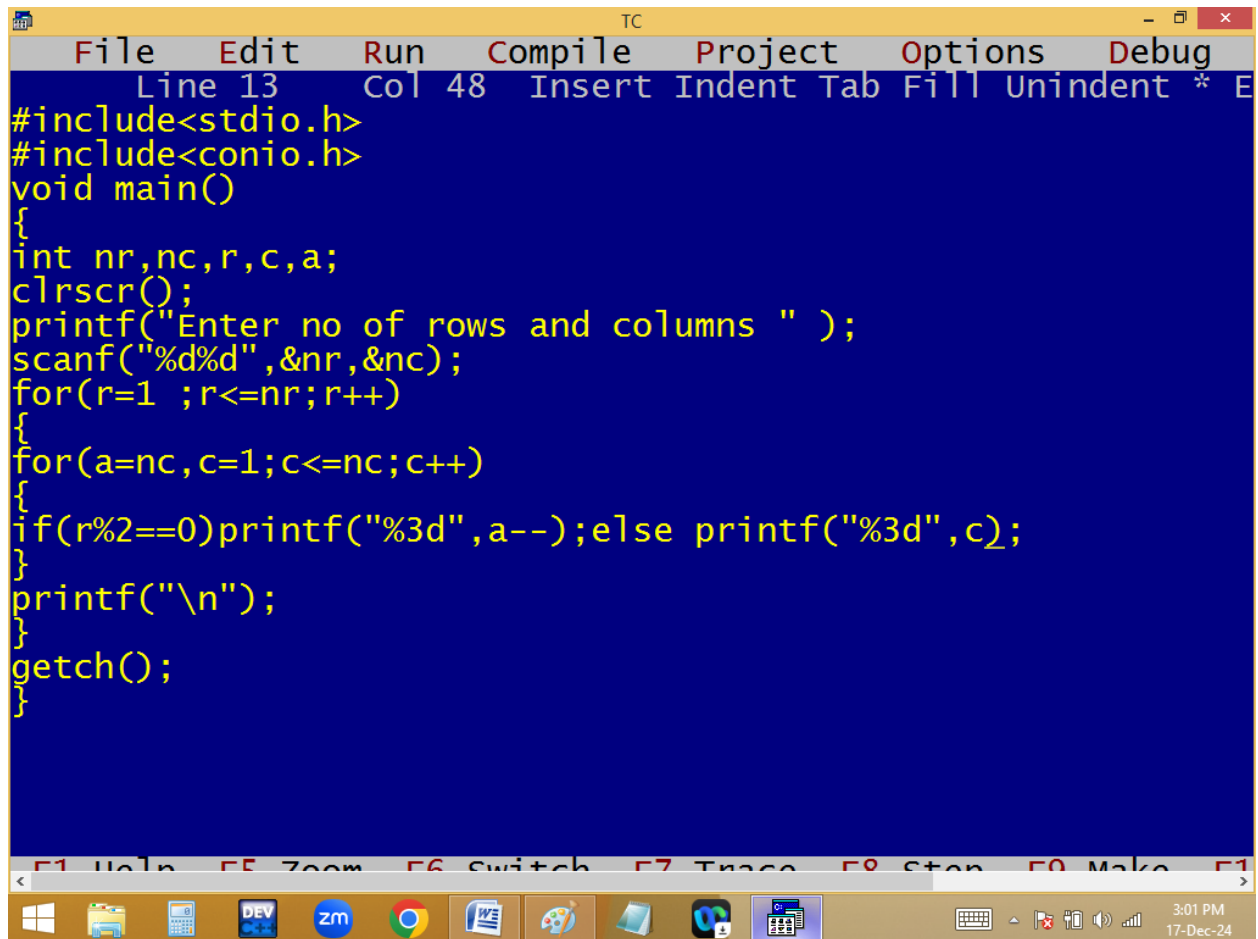
```

int a=1;
for( r=1; r<=4; r++)
{
    for( c=1; c<=4; c++)
    {
        p(r%2==0) p(--a);
        else p(a++);
    }
    p("\n");
}

```



$\frac{a}{1 \ 2 \ 3 \ 4 \ 5}$
 $4 \ 3 \ 2 \ 1$



```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 48 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(a=nc,c=1;c<=nc;c++)
{
if(r%2==0)printf("%3d",a--);else printf("%3d",c);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

3:01 PM
17-Dec-24

```

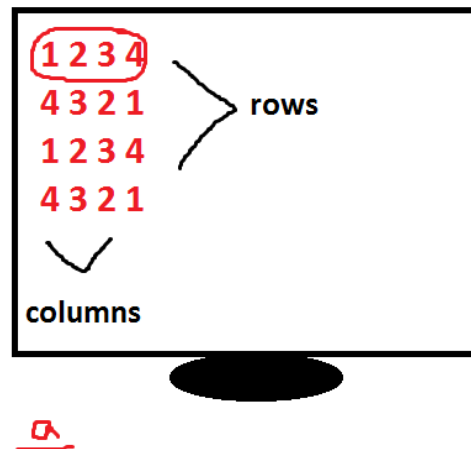
TC
Enter no of rows and columns 7 8
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8

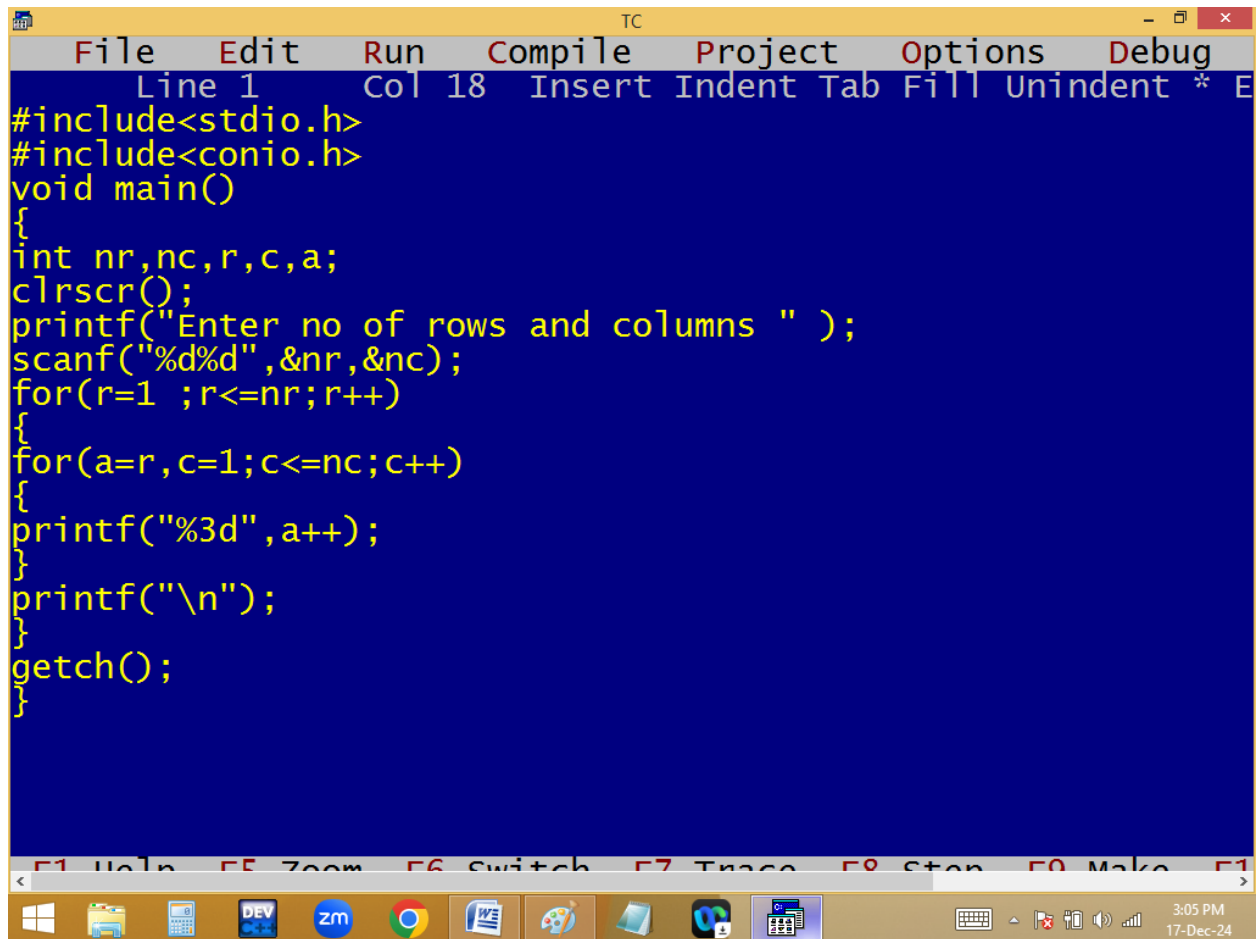
```

```

int a ;
for( r=1; r <= 3; r++)
{
    a=nc;
    for( c=1; c <= 4; c++)
    {
        p(r%2==0) p(a--);
        else p( c );
    }
    p("\n");
}

```





The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Edit" menu, there is a submenu with the following options: Line 1, Col 18, Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a dark blue background with yellow text. The code is as follows:

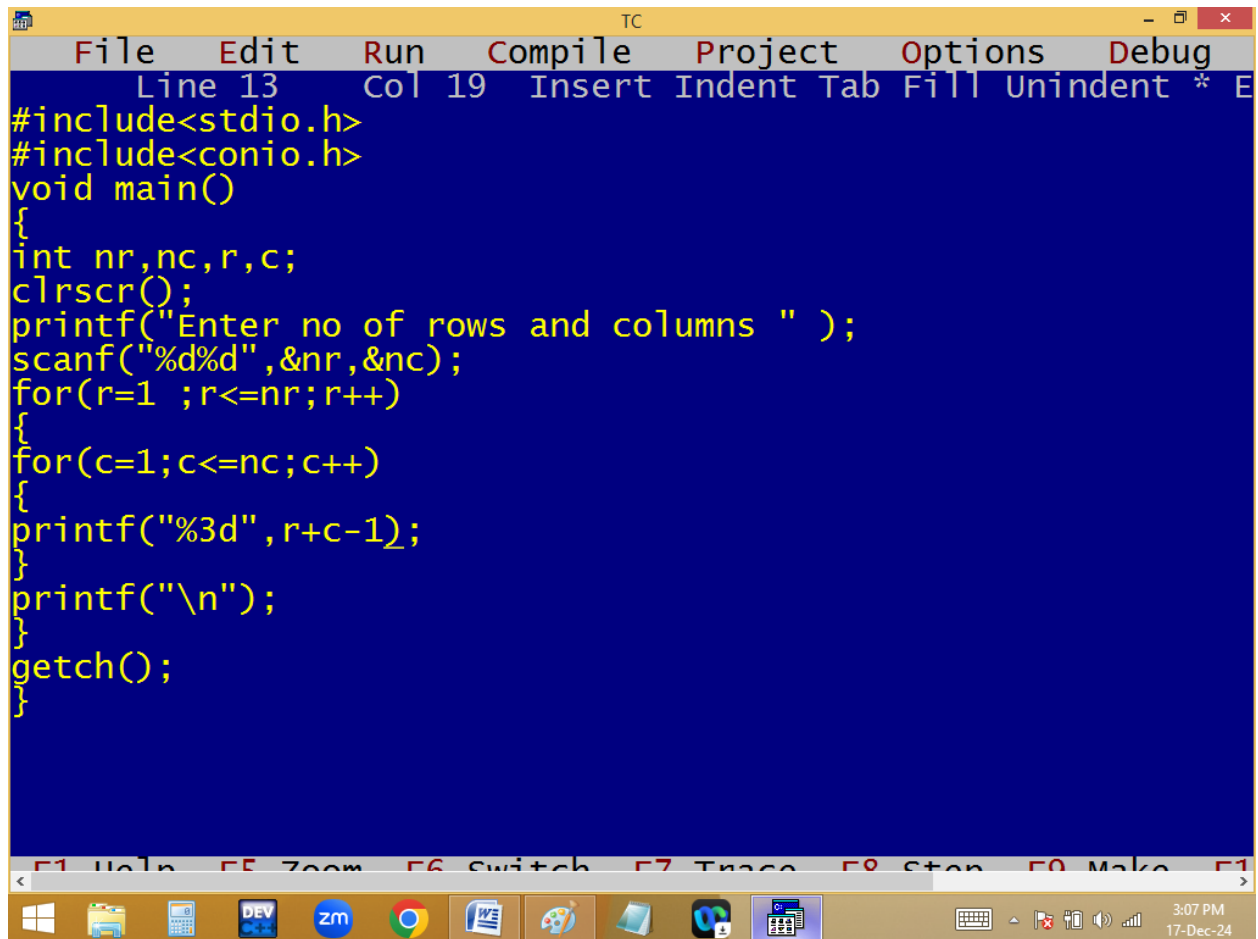
```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(a=r,c=1;c<=nc;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a status bar with the following text: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the status bar is a Windows taskbar with various icons, including the Start button, File Explorer, Calculator, DEV, ZM, Chrome, Word, Paint, and a folder icon. The system clock in the bottom right corner shows 3:05 PM on 17-Dec-24.

```
TC
Enter no of rows and columns 5 9
1 2 3 4 5 6 7 8 9
2 3 4 5 6 7 8 9 10
3 4 5 6 7 8 9 10 11
4 5 6 7 8 9 10 11 12
5 6 7 8 9 10 11 12 13
```



```
TC
Enter no of rows and columns 10 5
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
6 7 8 9 10
7 8 9 10 11
8 9 10 11 12
9 10 11 12 13
10 11 12 13 14
```



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 19", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main text area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",r+c-1);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar on the right shows the time "3:07 PM" and the date "17-Dec-24".

```
TC
Enter no of rows and columns 5 5
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
```

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 21 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(c%2==0)printf("%2c",96+c);
else printf("%2c",64+c);
}

printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

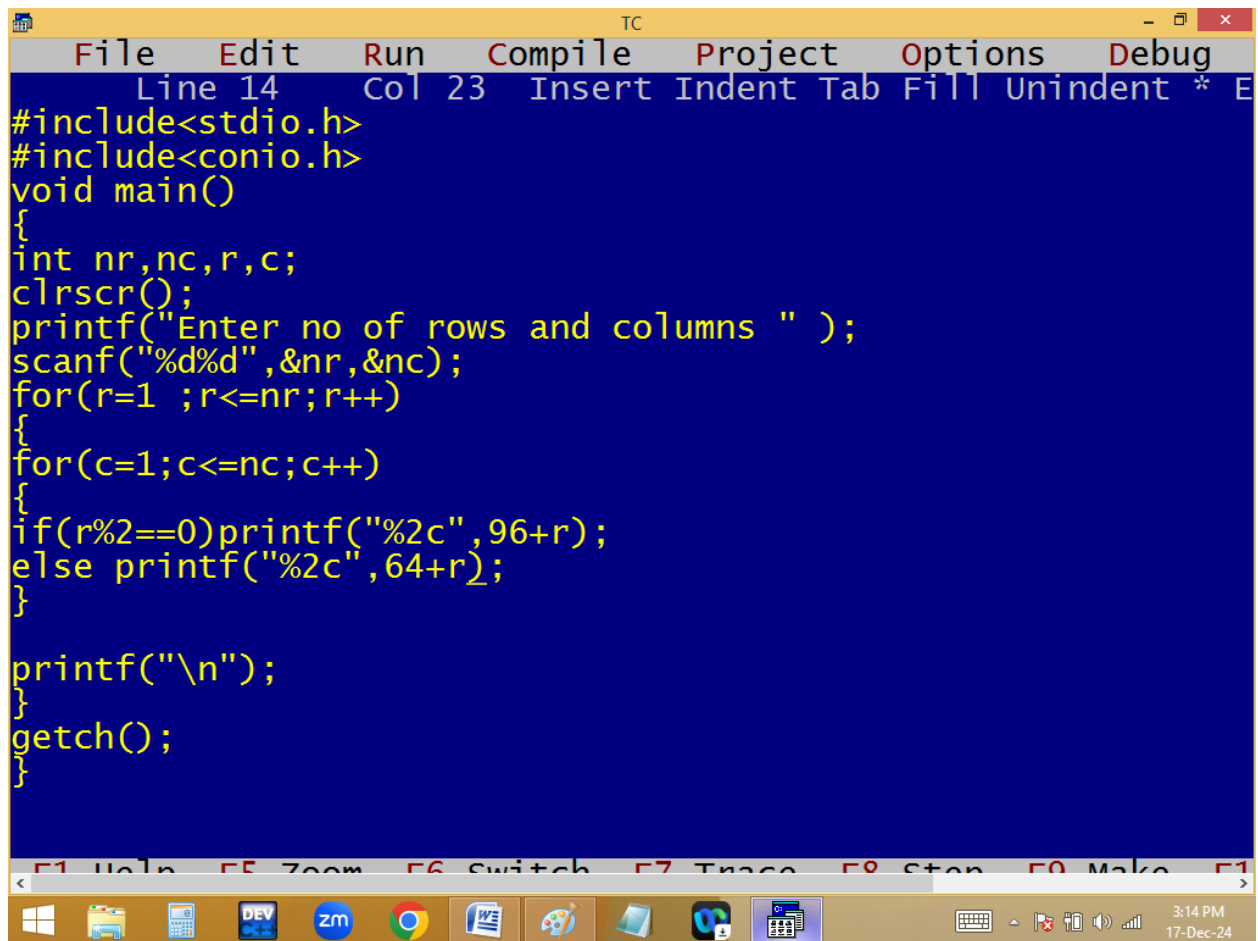
3:13 PM
17-Dec-24

TC

Enter no of rows and columns 10 26

A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z
A	b	C	d	E	f	G	h	I	j	K	l	M	n	O	p	Q	r	S	t	U	v	W	x	Y	z

3:13 PM
17-Dec-24



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 23", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text. It contains a C program that uses `stdio.h` and `conio.h` to print a checkerboard pattern of asterisks. The program uses nested loops for rows and columns, and conditional formatting to alternate between two different column offsets (96 and 64) based on the row number. The bottom status bar shows function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock shows "3:14 PM" and "17-Dec-24".

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",96+r);
else printf("%2c",64+r);
}

printf("\n");
}
getch();
}
```

```
TC
Enter no of rows and columns 10 10
A A A A A A A A A A
b b b b b b b b b b
C C C C C C C C C C
d d d d d d d d d d
E E E E E E E E E E
f f f f f f f f f f
G G G G G G G G G G
h h h h h h h h h h
I I I I I I I I I I
j j j j j j j j j j
```

```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 38 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;char L='a', U='A';
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",L);
else printf("%2c",U);
}
printf("\n"); if(r%2==0)L++;else U++;
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

3:31 PM
17-Dec-24


```

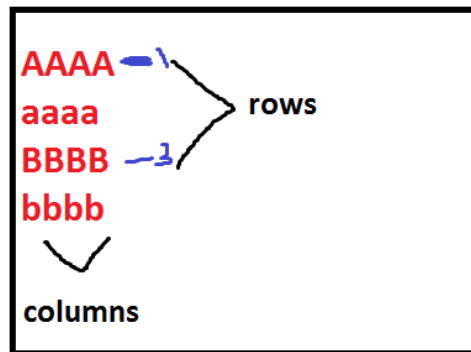
Enter no of rows and columns 20 10
A A A A A A A A A A
a a a a a a a a a a
B B B B B B B B B B
b b b b b b b b b b
C C C C C C C C C C
c c c c c c c c c c
D D D D D D D D D D
d d d d d d d d d d
E E E E E E E E E E
e e e e e e e e e e
F F F F F F F F F F
f f f f f f f f f f
G G G G G G G G G G
g g g g g g g g g g
H H H H H H H H H H
h h h h h h h h h h
I I I I I I I I I I
i i i i i i i i i i
J J J J J J J J J J
j j j j j j j j j j

```

```

char L='a', U='A';
for( r=1; r<=3; r++)
{
    for( c=1; c<=4; c++)
    {
        if(r%2==0)p(L);
        else p(U);
    }
    p("\n"); if(r%2==0)L++;
    else U++;
}

```



a $\frac{Y}{1}$ A A A A $\frac{U}{A}$ L
 2 a a a a a
 3 B B B B b
 4 b b b b C

* * * *

* 1 2 *

* 3 4 *

* * * *

1 1 1 1

1 1 2 2

1 3 3 3

4 4 4 4

1 2 2 2

0 1 2 2

0 0 1 2

0 0 0 1

